SHYAM SHARMA

Applied AI Engineer | LLM Products, RAG, OCR & Production Machine Learning
shyamsharma3.1415@gmail.com | +91 6003643630 | linkedin.com/in/shyam-sharma-5bb919343 | github.com/Shyamistic

SUMMARY

Engineer who ships AI into real products and measures whether it works: OCR and semantic-search systems for a global non-profit serving 100+ centres, a graph-neural-network climate forecaster (ISRO national finalist) reported honestly against baselines, an AI storytelling app on AWS Bedrock, and an LLM incident-response system with a safety validator. Production SaaS experience in a regulated compliance product.

HIGHLIGHTS

ISRO BAH 2026 National Top 9 (Vayu) | **IEEE COMPSAC 2026** accepted paper | **AWS Global Vibe 2025 Winner** (AURA) | **100+ centres** served by the VRI inventory system | **6.56M-parameter model** trained on 45 years of data

EXPERIENCE

Software Engineer, Wordwatch (UK enterprise compliance SaaS) | Remote | Jun 2026 - Oct 2026

- Shipped 90+ pull requests and 150+ commits across 4 repositories (React/TypeScript SPA, C#/.NET services) in a Jira-tracked agile team; owned features from API to UI.
- Built automatic risk-flag detection on transcription, IM and SMS ingest, and enterprise access control (RiskFlag / RiskFlagEdit) across the SPA and services.
- Contributed to the monitoring and observability stack (Prometheus, Grafana, Alertmanager, Elasticsearch, RabbitMQ) and fixed CI/CD failures.

Software Engineer (Volunteer), Vipassana Research Institute | Remote | Oct 2024 - Present

- Lead the VRI book inventory system (TypeScript, React, Supabase) used by 100+ meditation centres worldwide; architected its three-tier semantic search pipeline (stored QA, content, metadata) with confidence thresholds.
- Contributed to the team's OCR and metadata-extraction platform (Flask, React, MongoDB; six OCR engines, English and Hindi), the Pala platform (semantic-search RAG with embeddings, MCP-based storage with deduplication, RBAC) and audio fingerprinting (Chromaprint, CLAP embeddings, Milvus).

AURA: Autonomous SRE Platform | AWS Global Vibe AI Coding Hackathon 2025 Winner | Arcanum.AI Mentorship Program

- Designed sandbox-first remediation: every AI-generated fix is validated in an isolated environment before it reaches production (AWS Bedrock, Amazon Q Developer), built to cut incident recovery time.
- Selected for the Arcanum.AI mentorship program, with $10,000 of productionization support.

PROJECTS

Vayu: AI Digital Climate Twin of India | ISRO BAH 2026 National Top 9
Code: github.com/Shyamistic/vayu

- Trained a 6.56M-parameter GraphSAGE + Transformer model (PyTorch) on 45 years of data from 6 sources for 7-day rainfall and temperature forecasts; reported every score beside a tuned climatology/persistence baseline, beating it for rainfall in 3 of 4 regions on held-out 2023-2025 data.
- Found and fixed a silent failure where a stale model file returned plausible output; shipped the FastAPI service (32 routes), a React/CesiumJS interface and Kalman-filter assimilation of live sensor data, with 400+ backend and 1,600+ frontend tests.

Nexus Prime: Agentic AI for Incident Response (Azure) | IEEE COMPSAC 2026 (accepted) | Imagine Cup Regional Finalist
Live: white-river-06eae7700.2.azurestaticapps.net | Code: github.com/Shyamistic/Nexus-prime

- Evaluated an LLM root-cause system on 150 production incidents: 90% top-1 diagnostic accuracy; a safety validator cut hallucination rate from 8.0% to 1.2% and median diagnosis time fell from 48.3 min to 6.1 s.

Fablecraft: AI Storytelling App for Children (AWS) | H0 Hackathon
Live: fablecraft-pi.vercel.app

- Shipped a Next.js and FastAPI app on EC2 that turns a child's drawing into a narrated 8-scene story: Bedrock Nova for vision, story and image generation, Nova Lite for content moderation, Polly narration, S3 and single-table DynamoDB.

Finsa AI: Multi-Agent Loan Origination | SBI Hackathon 2026
Code: github.com/Shyamistic/Finsa-ai

- Built a speech-to-offer pipeline (browser speech, Bedrock entity extraction, fraud and credit scoring, policy-based offers) with a hash-chained audit log and 17 property-based tests.

A.S.T.R.A. (AI Kavach): Autonomous Cyber Reasoning System | Indian Army Terrier Cyber Quest 2026
Code: github.com/Shyamistic/A.S.T.R.A

- Built an offline vulnerability-repair system verified by replaying exploits in sandboxes; repaired two real CVEs (cJSON, zlib) and benchmarked on 104 CVE fix pairs.

SKILLS

Applied AI: LLM applications, RAG and embeddings, semantic search, OCR pipelines (Google Vision, Azure, Tesseract, EasyOCR), prompt engineering, evaluation against baselines, uncertainty estimation, audio fingerprinting
Models and platforms: PyTorch, AWS Bedrock (Nova), Azure OpenAI, Gemini, Ollama, FAISS, Milvus
Engineering: Python, TypeScript, SQL, FastAPI, Flask, React, Next.js, PostgreSQL, MongoDB, Redis, Docker
Cloud and DevOps: AWS, Azure, GCP, Git, GitHub Actions CI/CD, Prometheus, Grafana

EDUCATION & ACHIEVEMENTS

MCA, IIT Patna and IIIT Ranchi joint program | Jul 2025 - Oct 2027 | CGPA 9.80/10 | **BCA, Gauhati University** | Oct 2021 - Nov 2024
Achievements: Pravega Agile ML Challenge, IISc: National Rank 2 | GitLab i-Hack, IIT Bombay: Top 10 of 5,000+ | Microsoft Imagine Cup Regional Finalist | IMC Prosperity 4: World Rank 332 of 18,800+ | Published nova-architect on PyPI
Certifications: Oracle Certified: Generative AI Professional, AI Foundations Associate, DevOps Professional; OCI 2025